



**TAL
TECH**

e-GOVERNANCE LEGAL FRAMEWORK

ITE4170

Silvia Lips, Ph.D.
Norbert Nahayo Ndashimye, MSc
Tallinn University of Technology

04.09.2025

ABOUT ME

- Internal security background (CMB, SMIT, PBGB);
- More than 15 years of experience with Estonian eID;
- LL.M, MSc, Ph.D.;
- Currently lecturer and researcher at the Tallinn University of Technology and head of eID Competence Centre;
- Scientific expert at the University of Luxembourg;
- An eID expert and SPOC of eIDAS at the Information System Authority.

Interesting projects

- Estonian e-Passport tender (2006/2007);
- Estonian Mobile-ID (2015/2016/2022);
- Estonian eID card tenders 2015 and 2016, 2023;
- ROCA „crisis“ 2017;
- Estonian e-Passport tender 2019;
- Expert in the notification process of the eID schemes of the Netherlands, Denmark, Poland, Bulgaria, Norway, Czech Republic, Cyprus, France, Finland, Sweden etc.
- eIDAS v. 2. discussions,
- Estonian Identity Management Strategy 2018 and 2024



REPUBLIC OF ESTONIA
INFORMATION SYSTEM AUTHORITY



UNIVERSITÉ DU
LUXEMBOURG

TAL
TECH

TALLINN UNIVERSITY OF TECHNOLOGY

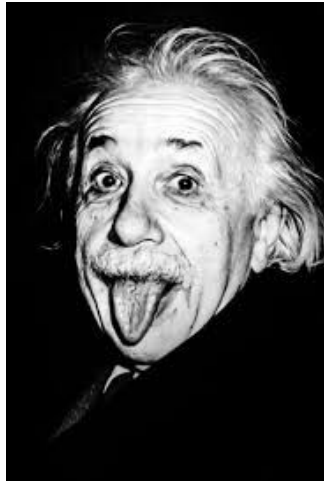


**TAL
TECH**

HOW ABOUT YOU?

- **Please introduce yourself briefly:**

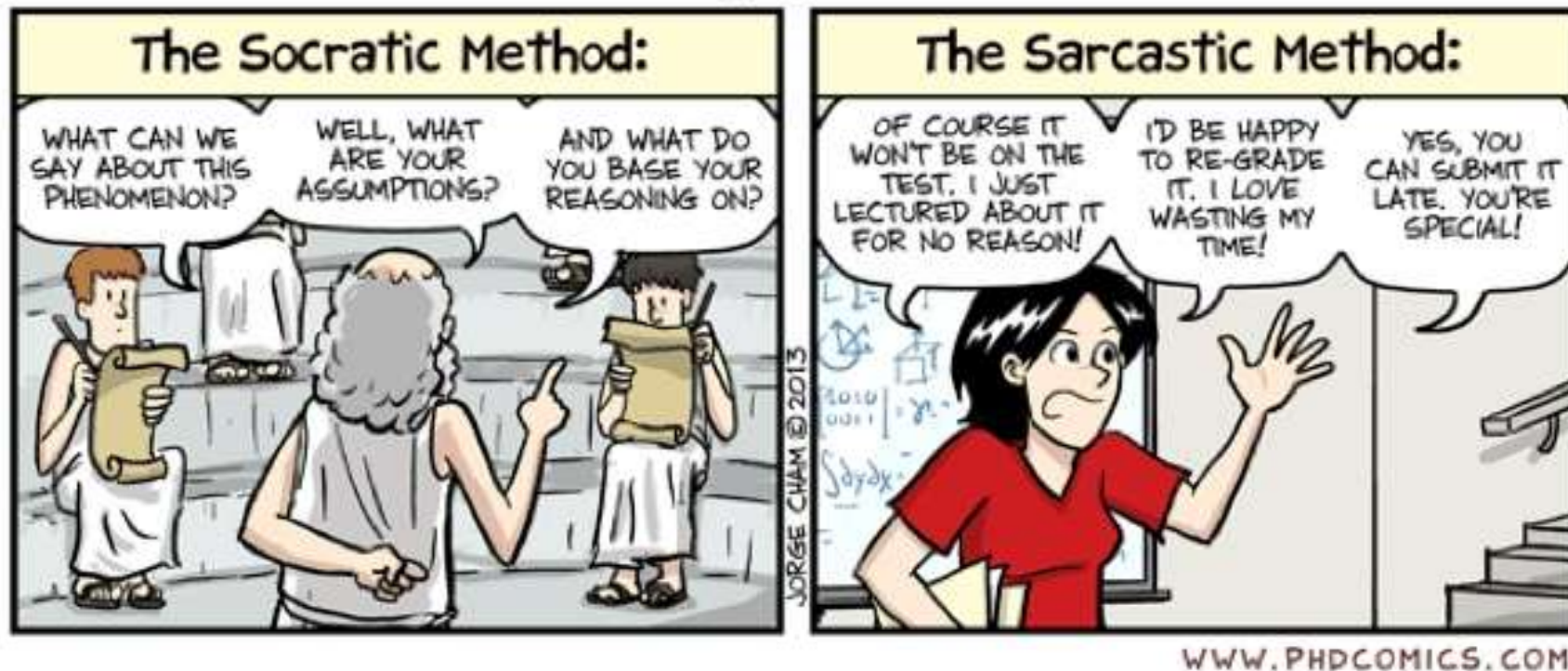
- Who you are?
- What is your background, and how do you evaluate your current level of legal knowledge?
- What are your expectations related to the course?



"Intellectual growth should commence at birth and cease only at death." Albert Einstein

TEACHING PHILOSOPHY

Teaching Methods



TEACHING PHILOSOPHY

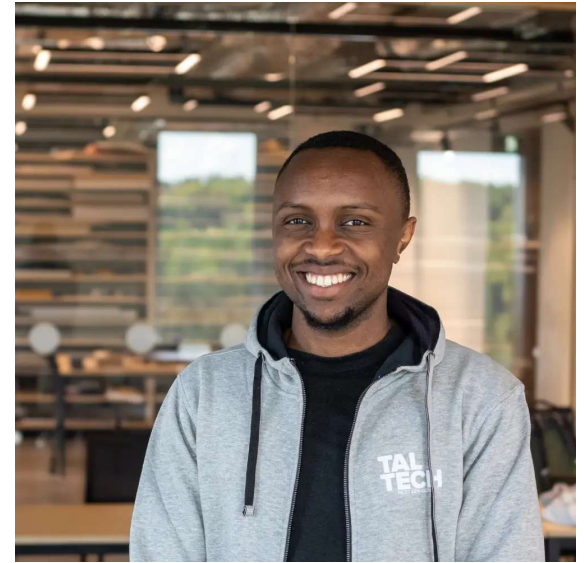
*„For me, life is a continuous learning and self-development path. I believe that **teaching is a bilateral process where all involved parties learn from each other.** It means that I see teaching as just one way of learning. Therefore, two-way communication and **dialogue between teachers and students is very important.** I see all students as **equal partners** who have different backgrounds, experiences, and personalities that makes interaction with them a valuable source of knowledge. I try my best to take into account the dynamics of every group and try to be adaptive and creative to encourage discussion, reflection, and critical thinking and enable successful knowledge transfer in a positive and stress-free environment.“*

OUTLINE

- **„e-Governance Legal Framework“ course overview**
 - Lectures and timetable
 - Assignments
 - Moodle
 - ...
- **Introduction to the e-governance legal framework**
 - General legal principles and pre-conditions for e-governance
 - E-governance legal framework and its components
 - Legal challenges
 - Discussion

GENERAL INFORMATION

- ITE4170 „e-Governance Legal Framework“
- The course will be assisted by Mr. Norbert Nahayo Ndashimye (Norbert.nahayo@taltech.ee)
- 6 EAP= 6x26h=156h, exam
- Lectures in September (04,05, 18,19), October (02,03) and November (05-08, hands-on week)
- Room SOC 415 or online
- **Moodle page:** <https://moodle.taltech.ee/course/view.php?id=35299>



TIMETABLE

- 04.09.2025 17:45-20:00 - **Introductory lecture** (Hybrid mode/SOC 415)
- 05.09.2025 16:45-20.30 - **EU regulatory framework for e-governance** (Hybrid mode/SOC 415)
- 18.09.2025 16:00-19:45 - **National legislation and e-governance** (NB! Online only)
- 19.09.2025 16:45-20:30- **Pre-juridical framework** (standards, guidelines, etc.) (NB! Online only)
- 02.10.2025 17:45-20:00 – **Presentations** (Room SOC 415)
- 03.10.2025 16:00-20:30 - **Presentations** (Room SOC 415)
- 05.11-08.11.2025 - **Hands-on training week on spot**
- 19.12.2025 -Legal analysis submission deadline

ASSIGNMENTS

■ **Presentation**

- Analysing one EU or national-level legal act;
- Presentation for everyone ca 20 min (15 min presentation + 5 min questions and feedback)
- Detailed guidelines will be provided in a separate file in the Moodle.

■ **Legal analysis**

- Will be based on your selected case;
- Detailed guidelines will be provided in a separate file in the Moodle.
- Deadline 19.12.2025.

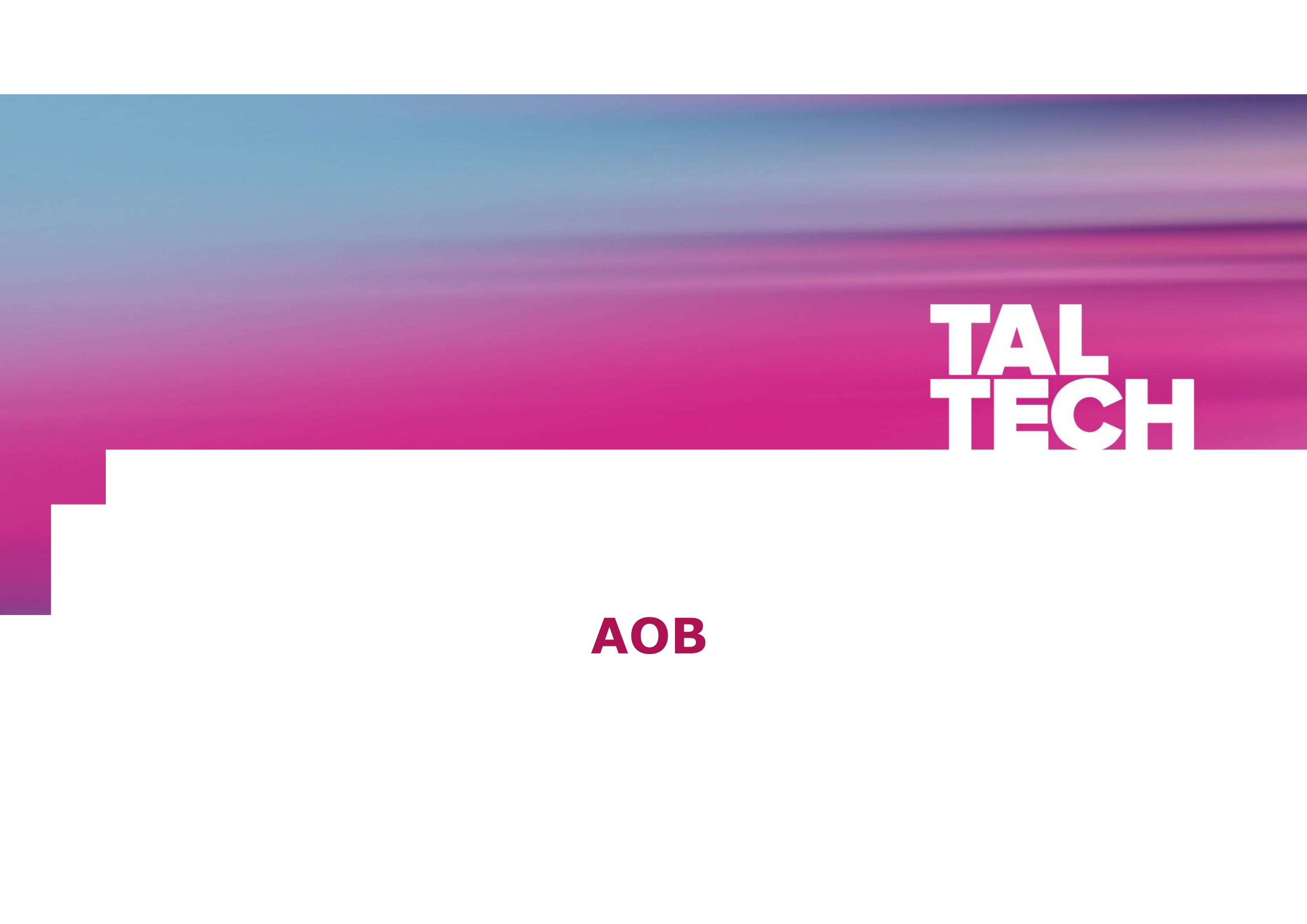
GRADING

- 40% (40 points) presentation and 60% (points) Legal analysis
- In total it is possible to collect 100 points.

- **Grading scale:**
 - 0–50 points – grade 0.
 - 51–60 points – grade 1.
 - 61–70 points – grade 2.
 - 71–80 points – grade 3.
 - 81–90 points – grade 4.
 - 91 points or more – grade 5

PREREQUISITES TO ATTEND

- Please be ready to use your **mobile phone and/or laptop** for seminar works.
- Try to find a spot for the lecture where you can focus and participate actively.
- Check the latest information at the Moodle page.
- Let's be open and reflective😊



**TAL
TECH**

AOB

**TAL
TECH**

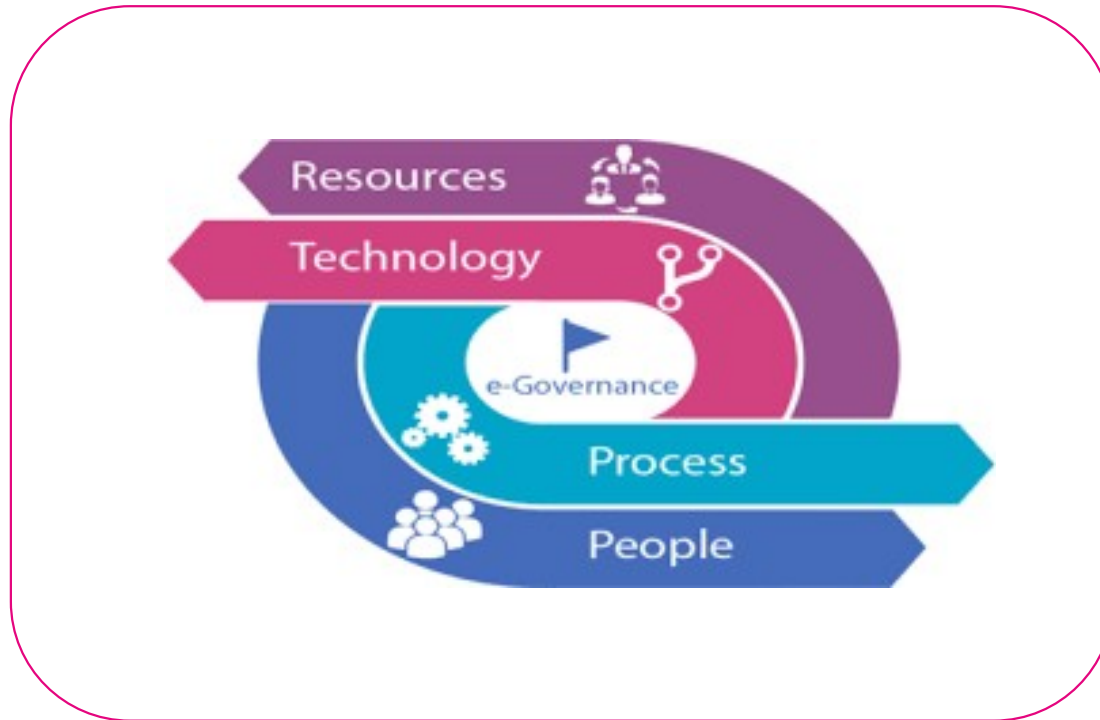
GENERAL LEGAL PRINCIPLES OF e- GOVERNANCE

HOW DOES SOMETHING BECOME A LAW?

- It all begins from the need
- Expert discussions together with proposal and drafting
- Introduction to the Legislative Authority
- Political debate and policy-making procedure
- Enactment (e.g. by president)
- Implementation and enforcement



LEGAL & REGULATORY FRAMEWORK OF e-GOVERNANCE



SOME BASIC PRINCIPLES WHILE REGULATING TECHNOLOGIES

IT is a science at the service of man:

- it must not infringe on human identity;
- the private life of the individual;
- human rights;
- individual or collective freedoms.



Isaac Asimov's Laws of Robotics

0. A robot may not harm humanity or, by inaction, allow humanity to come to harm.
1. A robot may not injure a human being or, through inaction, allow a human being to come to harm.
2. A robot must obey the orders given it by human beings except where such orders conflict with the First Law.
3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

PRE-CONDITIONS

- **Internet Access:**

- Access has to be real;
- Ensured practically and legally;
- Country specific aspects have to be considered;
- Affordable.
- E-governance legal environment should base on various legal acts (**not single regulation for everything**).
- General laws/area specific regulations.
- Important to find a balance between overregulation and a legal gap.
- Taking into account the **users (technology)readiness**.
- Conducting preliminary **legal environment analysis**



**TAL
TECH**

What kind of areas of laws are related to e-governance?

AREAS OF LAW INVOLVED

To avoid parallel systems, rules necessary for e-government should be regulated in legal acts.

- Legislation on Government structure and responsibilities, e-participation, e-voting;
- Access to Information legislation;
- Data Protection legislation;
- Legislation on digital signatures and Certification Services;
- ICT (electronic communications) legislation;
- Legislation on electronic commerce (and information society services), including consumer protection;
- Competition law, public Procurement legislation;
- Intellectual property legislation;
- Fight against cybercrime (Criminal law).
- Public procurement law.

LAW AND E-GOVERNANCE MATURITY MEASUREMENT

- **For example, the United Nations e-Government Survey 2024**
(Accelerating Digital Transformation for Sustainable Development - With the addendum on Artificial Intelligence)
- EGDI – e-Government Development Index
- LOSI – Local Online Service Index
- **Areas involved:**
 - Institutional framework
 - Services provision
 - Content provision
 - Technology
 - Participation and engagement
 - E-government literacy



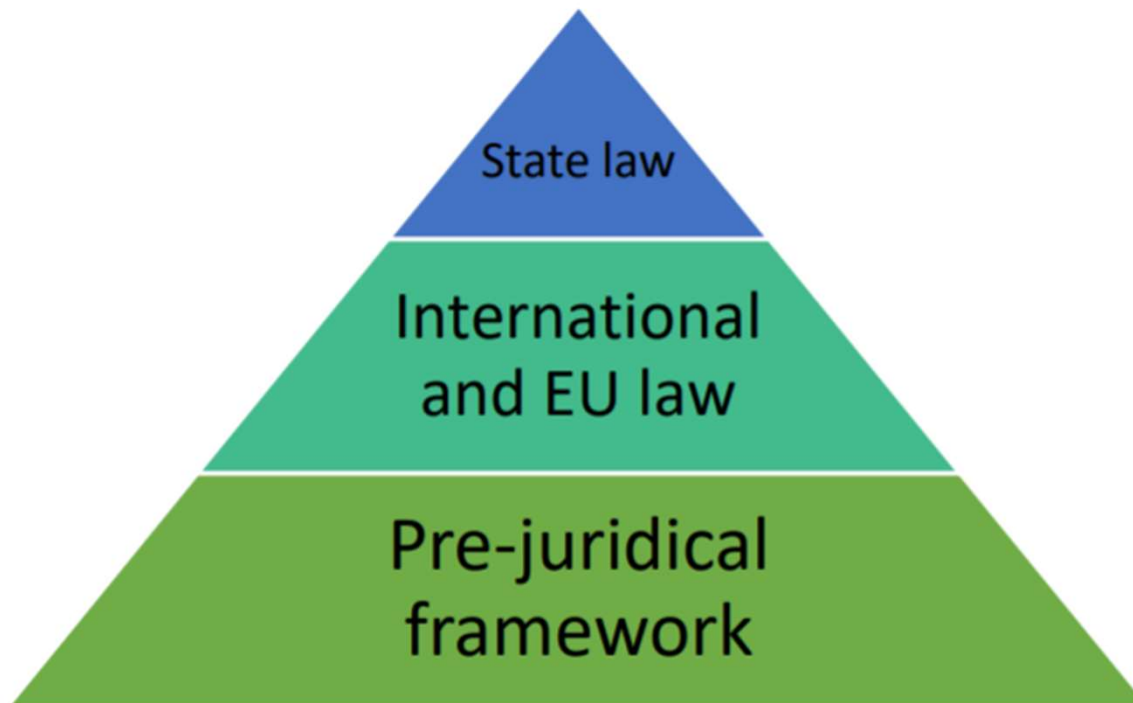
Table 4.1 Cities in the very high LOSI category, 2024

City	Country	LOSI value	City	Country	LOSI value
Tallinn	Estonia	0.9271	Paris	France	0.8125
Madrid	Spain	0.9271	Reykjavik	Iceland	0.8125
Riyadh	Saudi Arabia	0.9167	Rome	Italy	0.8125
Copenhagen	Denmark	0.9063	Riga	Latvia	0.8125
Dubai	United Arab Emirates	0.9063	Zurich	Switzerland	0.8125
New York	United States of America	0.9063	Buenos Aires	Argentina	0.8021
Istanbul	Türkiye	0.8958	Zagreb	Croatia	0.8021
Berlin	Germany	0.8854	Almaty	Kazakhstan	0.8021
Seoul	Republic of Korea	0.8750	Auckland	New Zealand	0.8021
Singapore	Singapore	0.8750	Stockholm	Sweden	0.8021
London	United Kingdom of Great Britain and Northern Ireland	0.8750	Sofia	Bulgaria	0.7917
Shanghai	China	0.8646	Toronto	Canada	0.7917
Manama	Bahrain	0.8542	Doha	Qatar	0.7917
Tokyo	Japan	0.8542	Amsterdam	Netherlands (Kingdom of the)	0.7813
Kyiv	Ukraine	0.8542	Oslo	Norway	0.7813
Vienna	Austria	0.8438	Sydney	Australia	0.7708
Bogota	Colombia	0.8438	Warsaw	Poland	0.7708
Moscow	Russian Federation	0.8438	Vilnius	Lithuania	0.7604
Sao Paulo	Brazil	0.8333	Guayaquil	Ecuador	0.7500
Montevideo	Uruguay	0.8333	Tel Aviv	Israel	0.7500
Helsinki	Finland	0.8125	Luxembourg-Ville	Luxembourg	0.7500

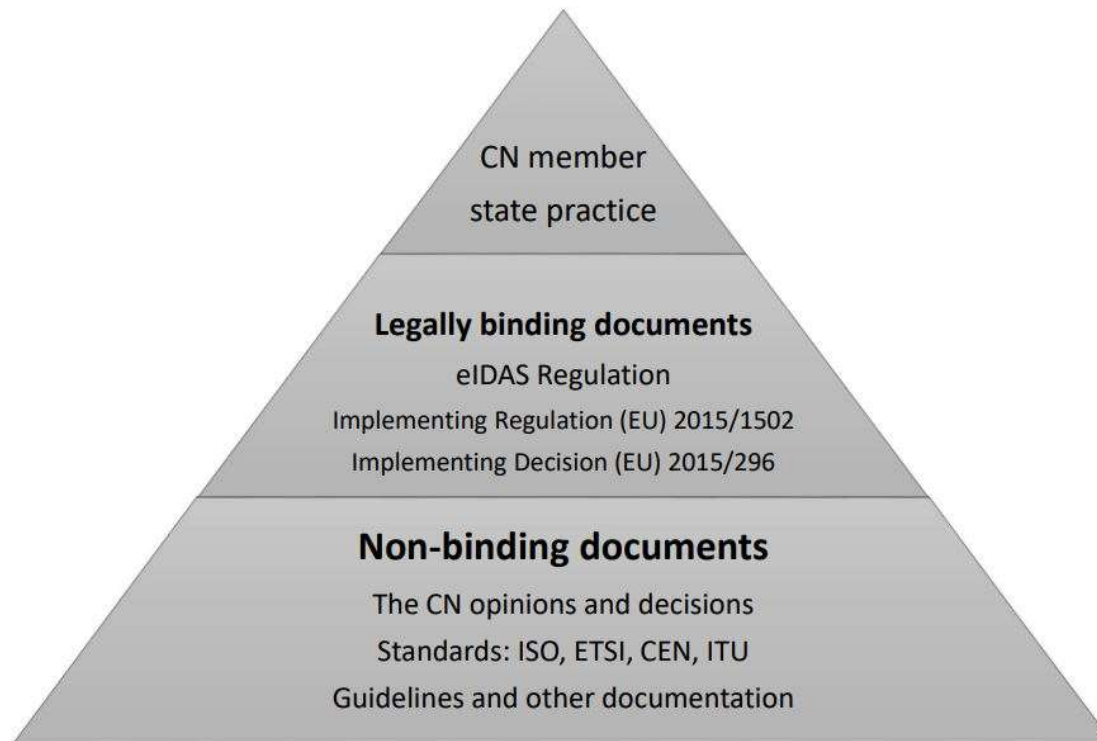
LEGISLATIVE COMPONENTS

- Cybersecurity
- Personal data protection
- National data strategy or policy
- Freedom of or access to information act
- Privacy statement on government websites
- E-government or digital government
- Digital ID
- Open government data
- Misinformation, disinformation or fake news
- E-participation
- AI
- Cloud

LEGAL AND REGULATORY FRAMEWORK



EXAMPLE FROM THE eID FIELD



PRE-JURIDICAL FRAMEWORK (SOFT LAW)

- International standards, guidelines, recommendations, strategies etc.

For example:

- ISO standards (ISO27001 – Information Security Management System; ISO/IEC 15408 - Common Criteria for Information Technology Security Evaluation (also known as CC), ISO9001- quality management etc.
- ICAO guidelines – Doc 9303 Machine Readable Travel Documents (passports, visas, ID-cards)
- European Standards Organisation (ETSI) - telecommunications, broadcasting and other electronic communications networks and services. TC ESI is responsible for Electronic Signatures and Infrastructures standardization within ETSI: <https://portal.etsi.org/TB-SiteMap/esi/esi-activities>
- Data Exchange standardization;

INTERNATIONAL AND EU LAW

- The Regulation (EU) N°910/2014 on **electronic identification and trust services** for electronic transactions in the internal market (eIDAS)
- Regulation (EU) 2024/1183 of the European Parliament and of the Council of 11 April 2024 amending Regulation (EU) No 910/2014 as regards **establishing the European Digital Identity Framework** (eIDAS 2.0)
- REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on contestable and fair markets in the digital sector (**Digital Markets Act**)
- Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on a Single Market For Digital Services (**Digital Services Act**) and amending Directive 2000/31/EC
- The General **Data Protection** Regulation (EU) 2016/679 (GDPR);
- Regulation (EU) 2015/2120 of the European Parliament and of the Council of 25 November 2015 laying down measures concerning **open internet Access**;
- Regulation (EU) 2019/881 of the European Parliament and of the Council of 17 April 2019 on ENISA (the European Union Agency for Cybersecurity) and on information and communications technology cybersecurity certification and repealing Regulation (EU) No 526/2013 (**Cybersecurity Act**)
- Decision (EU) 2015/2240 of the European Parliament and of the Council of 25 November 2015 establishing a programme **on interoperability solutions and common frameworks for European public administrations, businesses and citizens** (ISA2 programme);
- Directive 2014/24/EU of the European Parliament and of the Council of 26 February 2014 on **public procurement**.

NATIONAL LEGISLATION EXAMPLE

Estonian legislation:

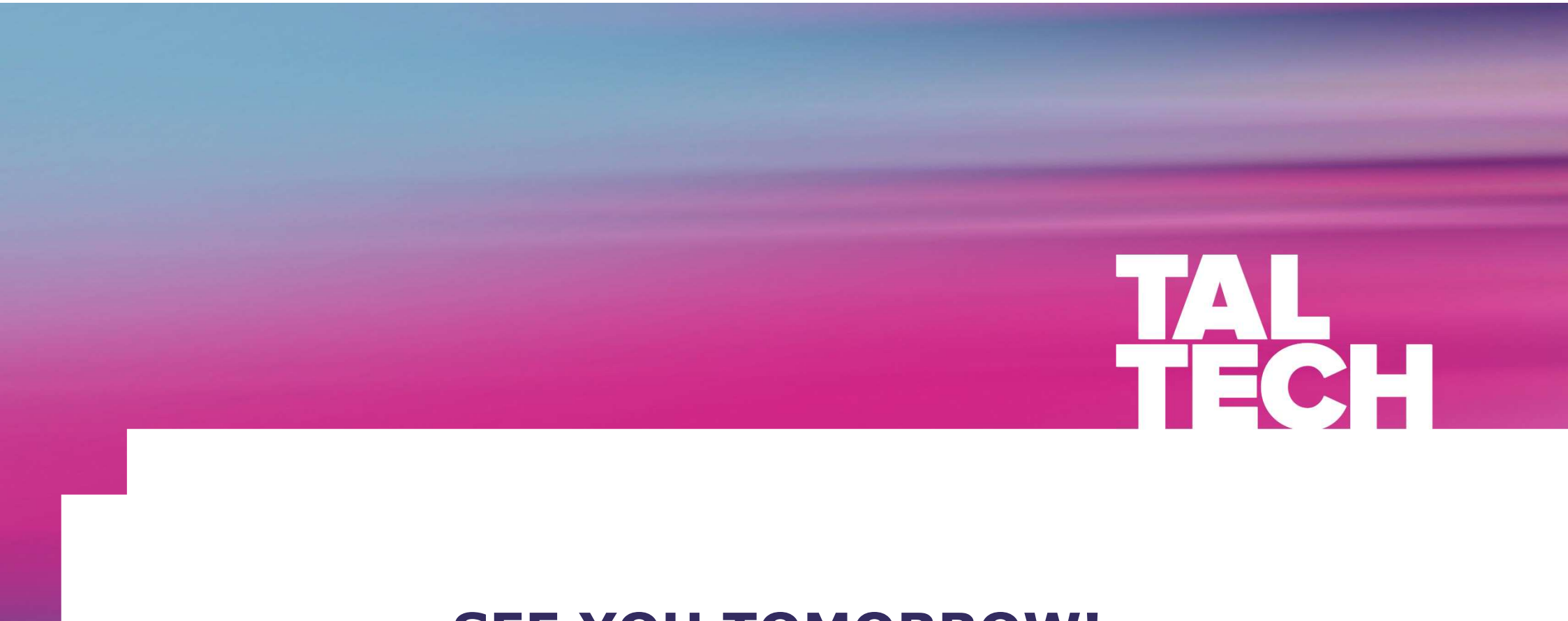
- Electronic Identification and Trust Services for Electronic Transactions Act (2016);
- Personal Data Protection Act (2018);
- Public Information Act (2001);
- Identity Documents Act (1999);
- Electronic Communications Act (2004);
- Competition Act (2001);
- Copyright Act (1992);
- Emergency Act (2017);
- Cybersecurity Act (2018);
-

The TAL TECH logo is positioned in the upper right corner. It features the words 'TAL' and 'TECH' stacked vertically in a bold, white, sans-serif font. The background behind the logo is a horizontal gradient transitioning from light blue on the left to a vibrant magenta on the right. A solid magenta rectangular block is located directly beneath the 'TAL' portion of the logo.

**TAL
TECH**

THANK YOU VERY MUCH!

Silvia Lips
e-mail: silvia.lips@taltech.ee



**TAL
TECH**

SEE YOU TOMORROW!